From the

ANNALS OF THE WHITE TOWER SOCIETY

For the Year C.R. 1286



OF THE SUPPOSITIONS ON WHICH THE GEOMETRY OF PLANES RESTS.

By Fairfax, of White Tower, Fellow of the White Tower Society.

WHITE TOWER: PRINTED FOR THE SOCIETY
C.R. 1286

IX.

OF THE SUPPOSITIONS ON WHICH THE GEOMETRY OF PLANES RESTS.

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Read before the Society in the Month of Reawakening, C.R. 1283; printed in the Annals for C.R. 1286.

INTRODUCTION

What the study of planes has done until now comes, for the most part, to two things: it has made registers of the planes, setting down where each meets the Main Material World; and it has kept their gates and their seals, that they may go on meeting it as they do. When Yardley made his Register of Planes he set down what had been seen and heard of the inside of each plane, but no measures; and the craftsmen who take measurements in the White Tower and in the demiplanes use the same geometry that they use in the cities. Whether that geometry holds in every plane, no one has asked.

Yet two things are already known that will not let the question rest. Severin's Third Question tells of demiplanes in which a man who walks straight on comes back at length to where he started. The trial demiplane that Holbrook forged at Kanaich has a gate no higher than a man, and one who walks straight on within it must go sixty paces before he comes back to the gate. A geometry that runs on flat without end, as that of the cities does, has no room for such a thing: by it, to walk straight on is only to go farther and farther away. It is plain, then, that the geometry within a plane can neither be seen from without nor be carried over from the cities, and that we must first ask on what it rests.

What I have to say is just this: on what suppositions the geometry of a plane rests; which of them belong to every plane, and which must be measured within each. I shall go in three steps: first, what a plane is, taken as an extension; secondly, what measure such an extension may carry; and lastly, how the measure of a plane is to be fixed from facts measured in it, and by what that measure is held. Many of you who hear me are law-mages and sealers and not reckoners, and I shall use as little reckoning as I can; the few places where I cannot avoid it are worked out in a note, which is kept by the Society.

I. EXTENSION OF MANY FOLDS

1. The discrete and the continuous. There are two kinds of how much in the world. One is counted piece by piece, as a box of crystals or a file of men, and this I call discrete; the other runs on from place to place, as position, the hour, or the affinity of a material, and this I call continuous. How much there is of the discrete is settled by counting; how much there is of the continuous must be found by laying something against it, and that is measuring. For discrete things the means of comparing is given already, for counting is comparing; for continuous things, what to lay against them has to be found besides. Metcalfe's tetrahedron has both: the affinities of natural crystals of a single form fall only upon his points, and can be counted one by one; the tetrahedron itself is continuous,

and every point of it is a possible affinity. To this difference between the discrete and the continuous I shall come back in the third part.

2. From one fold to many. If in some continuous extension one can pass from place to place only along one way, forward or back, I call it of one fold. The hour is of one fold. By the accounts of the elemental mages, the Grey Reach has only time and direction; if that direction is a single way to go, it too is likely of one fold. Let an extension of one fold be moved as a whole into places where it was not before, and what it sweeps is of two folds; let one of two folds be moved in the same way, and what it sweeps is of three; and so there may be extensions of any number of folds, as in Fig. 1. Conversely, if in an extension of many folds one fixes some quantity that varies continuously and stands still in no small part of it, then all the places where it has the same value make together an extension of one fold fewer; and fixing one quantity after another in this way, every place in an extension of n folds may be fixed by n numbers.

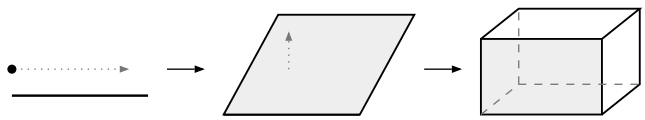


FIG. 1. From one fold to three: a point moving sweeps a line of one fold; the line moved as a whole into places where it was not sweeps a surface of two; the surface moved again sweeps a body of three (schematic)

The Main Material World in which we stand needs three numbers for a position, east and west, north and south, high and low, and is of three folds. The affinity of a material is a continuous extension as well: the four shares add up to one, so that when three are fixed the fourth is fixed also, and it is of three folds; Metcalfe's tetrahedron is a picture of it. A web-state at one place upon the Web of Laws is, as Elworthy describes it, the state of things at that place; I hold that it is not fixed until the rise and fall everywhere near that place is fixed, and such numbers cannot be counted out; the web-states taken together are an extension of folds without number. Extensions of many folds, then, are not space alone: whatever varies continuously and needs several numbers to be fixed is one.

3. Near at hand and as a whole. To fix positions by n numbers is a matter only of what lies near at hand about one place. How an extension is joined together as a whole these numbers do not tell. It is exactly the joining of the whole that Severin's Third Question asks after: two planes may each be like ordinary space of three folds near at hand, and yet not be of one shape as wholes. Of the joining of the whole I shall speak today only so far as the measure touches it, and leave the rest to the Third Question.

II. MEASURE

1. The rule that does not change. To compare lengths in a continuous extension one must first have a rule. The first and weightiest supposition is that a rule does not change wherever it is carried, so that the length of a piece of line depends on the piece alone, and not on where it lies, which way it points, or by which road it was brought there. The extension does not bring this with it; whether it holds in a plane must be measured. In what follows it is supposed to hold.

Some extensions have no rule ready to hand at all. How far the affinities of two materials lie apart may be measured by the error of the affinity meter, two affinities that differ by several times the error lying several times as far apart; or one may look only at how much each of the four shares differs, and take the greatest. Either can be defended, and which to take depends on what it is wanted for. We are so used to measuring the distances of space with a rule that we take measure to be something space brings with it; the example of affinity shows that measure is a further thing laid upon the extension, and that what is laid upon it needs a ground of its own.

2. The length of a small step. How long a very short step is, is fixed by how much each of the n numbers changes in taking it. The simplest way of fixing it is this: the square of the length of a small step is the sum of the products of the changes, taken two and two, each product with a coefficient of its own; and this sum is positive for every small step. These coefficients pass smoothly from one place to the places near it, and so do the rates at which they change, without sudden leaps: such an extension I suppose to be smooth everywhere. In flat space, with places numbered along three directions square to one another, this is the old rule of the right triangle, the square of the step being the sum of the squares of the three changes. Other ways there are: in the example of affinity above, to take the greatest of the four differences is not a sum of the second degree. But I can think of no plane that would need such a way, and I shall speak only of the simplest.

3. How many of the coefficients belong to the extension. The changes of n numbers taken two and two, each with itself as well, make $n(n+1)/2$ products, and so there are that many coefficients, each varying from place to place. But which n numbers we use to number the places is our own choosing, and when the numbering is changed, the coefficients change with it. What may be chosen at will in the numbering is n numbers varying from place to place; when they have been chosen, there remain $n(n-1)/2$ quantities varying from place to place, and these the extension itself fixes, as in Table I. At any single place, the numbering can always be chosen so that the coefficients there are those of flat space; what belongs to the extension itself must be sought in the way the coefficients change from place to place.

TABLE I. How many quantities varying from place to place fix the measure of an extension of n folds

Folds	*Coefficients*	*Taken away by the numbering*	*Fixed by the extension*
1	1	1	0
2	3	2	1
3	6	3	3
4	10	4	6
5	15	5	10

In an extension of one fold the extension fixes nothing: however a line is bent, one who measures it walking only along it cannot feel the bending. In one of two folds it fixes one quantity, and in one of three folds, three.

4. The bending. What it is that the extension fixes may be seen thus. At a place take two directions, and in every direction spread between them walk out from that place by the shortest way; these ways weave a small surface; how that surface bends I call the bending of the place in those two directions. One who measures it need not leave the surface. Let

him draw upon it a very small triangle whose three sides are all shortest ways: if its three angles are together more than two right angles, the bending is forward; if less, backward; and the part over or short, set against the area of the triangle, is the size of the bending. A place has a bending for every pair of directions; the pairs are without number, yet all these bendings are fixed by a few numbers: one at a place in two folds, six at a place in three. When the bendings at every place are known, the measure is fixed, however the numbering is changed; and in general it is enough to know how the bendings in $n(n-1)/2$ chosen directions of surface change from place to place, which is the number of the last column of Table I. An extension in which the bending is nought everywhere and in every direction is, near at hand, the ordinary flat extension, and its geometry near at hand is the geometry of the cities.

The bending here spoken of is the bending that one who lives within the extension can measure, and it is another thing from whether it looks bent from without. A sheet of paper rolled into a tube looks bent from without, and yet the angles of a triangle drawn upon it are still together two right angles, for the paper has been neither stretched nor shrunk; one who lived in the paper would not feel that it had been rolled. On the other hand, a surface like the bottom of a bowl cannot be laid flat without stretching or shrinking, however small a piece of it is cut, and one who lives in it can tell as much by measuring triangles. A plane has no without on which to stand; we can only live within it and measure, and so this bending alone is ours.

5. Even bending. An extension in which the bending at a place is the same in every direction, and the same at every place, I call evenly bent. Only in an evenly bent extension can a figure be moved to any place and turned any way without being stretched or shrunk. When a craftsman in the city carries a table or turns an array-stone and its shape stays as it was, he relies upon this, that the space of the city is evenly bent, and its bending too small to be measured. Near at hand there are only three kinds of evenly bent extension: the bending forward, nought, or backward. Fig. 2 shows triangles on surfaces of the three kinds.

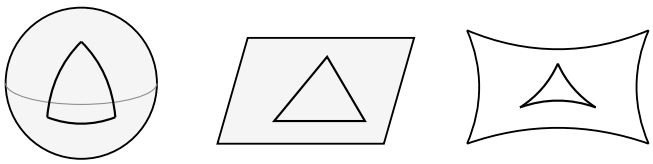


FIG. 2. Triangles whose three sides are all shortest ways, on evenly bent surfaces of the three kinds: bending forward (left), the angles more than two right angles; nought (middle), equal to them; backward (right), less (schematic)

In an evenly bent extension whose bending is forward, the three angles of a triangle are always more than two right angles, and the larger the triangle, the more they exceed. How much it is bent may also be told by a radius, after the sphere of the heavens of the makers of almanacs: the greater the bending, the smaller the radius. If within it one can always walk on as far as one will, the whole is finite, and yet has no edge: one who walks straight on comes back, after a certain way, to where he set out; when it is joined in the

simplest way, that way is 2π times the radius, and if each place is joined with the place right opposite it, only half as much. In one whose bending is backward, the three angles are always less than two right angles, and the whole may be without end. Where the bending is nought, the geometry near at hand is the ordinary one.

6. Without edge and without end. Two things are thus told apart which are often taken for one. To be without edge is to be such that wherever one comes, one may still go on; this belongs to the extension itself, and does not depend on the measure. To walk on as far as one will, and to be able to go infinitely far, belong to the measure. A plane of three folds that has no edge need not be without end: an evenly bent plane whose bending is forward, if one can walk on in it as far as one will, has no edge, and is finite. The demiplanes of which Severin speaks may be regarded so, or may be of the kind spoken of next.

One mistake must also be guarded against. To come back to where one started does not show that a plane is bent. A plane whose bending is nought everywhere may yet, if it is joined together differently as a whole, be finite and without edge, like a sheet of paper rolled into a tube with its two ends joined as well: one walks out through one wall and comes back in through the wall opposite. In such a plane one who walks straight on in certain directions also comes back to where he started, but the way back is of different lengths in different directions, and in many directions he never comes back at all; and the angles of a triangle are still together two right angles. Severin himself said that to come back to where one started is only one of the looks of such planes, and cannot be taken for a test. To know whether a plane is bent or only joined differently, then, it is not enough to see whether one comes back; one must measure triangles.

III. APPLICATION TO PLANES

1. What facts fix the measure. By what has been said, to fix the measure of a plane one must first try whether a rule changes when it is carried about, by any road; then measure triangles at every place and in every direction. If at one place triangles of different sizes each give, reckoned upon an evenly bent surface, the same radius (for very small triangles, the same proportion of their excess or shortfall to their area), the bending of that place in that surface has been measured; if this is so at every place and in every direction, and the radius is the same, the plane is evenly bent, and the size of its bending is known as well.

2. The trial demiplane at Kanaich. The gate of the trial demiplane that Holbrook forged at Kanaich is still kept by the White Tower Society. With two Fellows of the Society I have measured within it for seven years, stopping for one year in the tide year 1280. First the rules: two rules exchanged and compared at every place and in every direction within it, and carried to one place by roads of different length and compared again, agree everywhere; carried out through the gate and laid against the rules of the city, they have not changed their length. Then the triangles: six triangles of different sizes were taken, near the gate, in the middle and far within, some lying level and some upright, their three sides stretched with cord along the shortest ways and their three angles measured with the angle-meter, each three times and the mean taken, the error of an angle after the mean being within a tenth of a degree, as in Table II. In all six the three angles together exceed two right angles, and the larger the triangle, the more. Reckoning by an evenly bent surface whose bending is forward, which is the method the makers of almanacs use for triangles upon the sphere of the heavens, and working back from the three sides and

the sum of the angles of each triangle to the radius of its surface, all six come out between 9.5 and 9.7 paces, the mean 9.64 paces; the small triangles are hard to measure and their radii scatter more, the large scatter less, and weighting each triangle by its error gives 9.63 paces, with an error of some three hundredths of a pace. What was measured at different places and in different directions agrees.

TABLE II. Six triangles in the trial demiplane at Kanaich

Three sides (paces)	*Place and lie*	*Sum of the angles (degrees)*	*Excess (degrees)*	*Area reckoned as flat (square paces)*	*Radius worked back (paces)*
2, 2.5, 3	Gate, level	181.55	1.55	2.5	9.62
3, 3, 3	Gate, upright	182.49	2.49	3.9	9.53
4, 5, 6	Middle, level	186.24	6.24	9.9	9.71
5, 5, 5	Middle, upright	186.79	6.79	10.8	9.72
6, 7, 8	Far within, slanting	193.43	13.43	20.3	9.64
8, 8, 8	Far within, level	198.88	18.88	27.7	9.60

If the radius is 9.64 paces and the plane is joined in the simplest way, one who walks straight on should come back to where he started after 2π times the radius, that is, 60.6 paces. We set out from the gate in five directions, once in each, and in the five walks the way back to the gate was between 60.3 and 60.7 paces, the mean 60.5, which gives back a radius of 9.63 paces and agrees with the triangles: the triangles give the length of the way round, and the walks give the radius, and the two roads meet. The way back was of one length in all five directions, and not of lengths changing with the direction, as in a plane whose bending is nought and which is only joined differently; besides, in such a plane the angles of a triangle would be two right angles, and here all six exceed them. And the way was some sixty paces, not some thirty, so that the plane is not of the kind in which each place is joined with the place right opposite it. This trial demiplane, then, apart from the small place of its gate, is of three folds, evenly bent, its bending forward, finite as a whole and without edge: what Severin told of, that one walks on and comes back to where one started, is here a thing that can be measured and reckoned. Whether its joining as a whole might yet be of some other kind is to be told by Severin's own method of the cord. Fig. 3 draws the excess against the area of the triangles: the four small ones lie near a straight line, and the larger rise above it by degrees, because the area has been reckoned as flat; reckoned upon an evenly bent surface, all six agree.

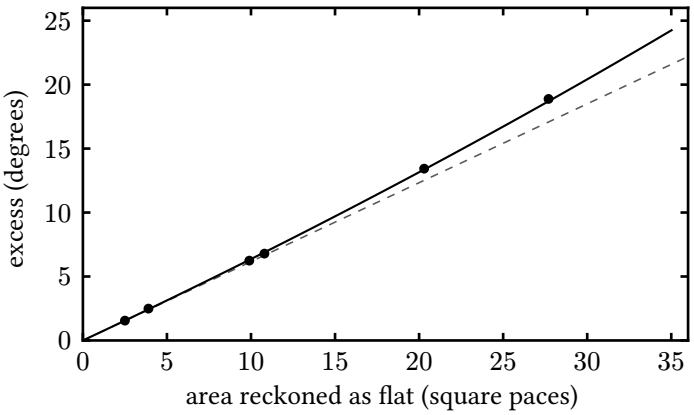


FIG. 3. The excess (degrees) of the six triangles in the trial demiplane at Kanaich against their area reckoned as flat (square paces): the points are those of Table II; the straight line is the proportion that very small triangles should give for a radius of 9.64 paces; the curve is reckoned for equal-sided triangles upon an evenly bent surface of radius 9.64 paces, and the three unequal-sided triangles of Table II, reckoned for the same radius, lie above it by no more than some 0.05 degree

3. The Main Material World. No craftsman of the cities has ever measured a triangle whose angles were not together two right angles; but the triangles they measure are too small. I asked the stations of the Tower of All Things to measure three great triangles for me, each made by three stations sighting one another from high places, by lamps at night, with sides from near a hundred to more than two hundred miles, as in Table III. Here the sides are lines of sight, and that light goes by the shortest way is another supposition; the air near the ground bends the line of sight, and so each two stations sighted one another both ways and the mean was taken; the angles were measured in the surface that the three lamps span. In each the sum of the angles differs from two right angles by a few seconds at most, within the error of measurement. That error is some six seconds; the difference measured in each triangle, with these six seconds added, is the most its excess or shortfall can be, and the greatest of these is some nine seconds. By the bound so found for each triangle, if the surface in which the three triangles lie is bent at all, its radius is at least thirteen thousand miles, and the largest triangle sets the bound at twenty-seven thousand five hundred.

TABLE III. Three great triangles measured by the stations of the Tower of All Things (error of measurement some six seconds)

Three sides (miles)	*Area reckoned as flat (square miles)*	*Sum of the angles less two right angles (seconds)*	*Radius of bending at least (miles)*
118, 143, 167	8,280	−1.2	15,400
204, 231, 265	22,735	−0.2	27,500
96, 158, 177	7,550	+3.1	13,100

Within the reach of our measuring, therefore, the level surface of the Main Material World is flat; for upright surfaces we have only the small triangles of the craftsmen of the cities, and the bound is much weaker. And this says only that its bending is too small for us to

measure, not that it is nought; whether the whole world has an edge, and how great it is, these triangles cannot tell.

4. The very great and the very small. That the measure is so within the reach of our measuring cannot be carried into what lies beyond it. Of the very great I have spoken. Of the very small it is harder still to speak: rules and triangles may be of no use in the very small. If a plane, in its smallest parts, is not continuous but woven strand by strand of lines of law, then its measure may be fixed by counting lines: how far two places lie apart is how many strands lie along the shortest string of lines between them, and the means of comparing is within the plane itself, as with the discrete things of the first part. If it is continuous down to its smallest parts, a rule cannot be had from it alone; something else must give it one.

5. By what the measure is held. Two things are known to you all. The floors of the White Tower can be widened, folded and closed by the governing of the laws; the same floor stays the same size seen from without, and yet the distances within are changed, and a flight of stairs does not always lead to what lies straight above. Holbrook stopped the keeping of the gate of one of his trial demiplanes, and in a few months the gate slowly closed, and what had been left within could never be taken out again. These two show that in a forged plane the measure, together with the way it is joined to what is without, is held by the laws and can be changed; in a forged plane, the rule is given by the laws. Whether it is so in other planes must be measured apart. Willoughby, who keeps the record of the running of the laws upon the gates of the White Tower, has found it all but unchanged for more than twenty years; if the measure of the Main Material World is held by laws too, those laws are steady.

[The passage that followed, concerning the folding in the depths of the White Tower, is not printed, by the Society's rules for sealed matters.]

To ask, then, why the measure of a plane is as it is, is at this point no longer a matter for reckoning. Which lines of law hold a plane together, and how they fix the bending at each place, is to be handed to the study of laws.

CONCLUSION

What I have said today only makes ready a manner of speaking for the study of planes: of how many folds a plane is; by what quantities its measure near at hand is fixed; which of them can be measured, and how. Of which kind a given plane is, one must go in and measure; it cannot be thought out sitting without. The geometry of the cities was measured in ordinary places; before it is carried into another plane, it must be measured there first.